Healthcare Analysis

Check out the project online!

The Outliers Team

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Overview

The goal behind this project is to provide a comprehensive analysis of healthcare datasets, giving an example of the correct approach to tackling this type of dataset.

The project will go through four phases: Cleaning, Modeling, Visualization and then Revision.

In the Cleaning phase, we will clean the dataset to make it ready for analysis.

In the Modeling phase, the cleaned dataset will be split into entity tables and the relationships between all of these tables will be determined, then finally we will create DAX measures for data insights.

In the Visualization phase, the previously calculated DAX measures will be used to create a beautiful report on the dataset

Finally, in the Revision phase, we will go over any ultimate changes that need to be made to the dataset

Dataset

Click here

SYNTHEA

SyntheticMass is a synthetic healthcare dataset modeling over one million fictional residents of Massachusetts, complete with realistic, but artificial health records generated by the Synthea tool. It provides a privacy-safe sandbox free of real PHI or PII for testing health IT applications, data analysis, and research.

The dataset consists of 18 tables covering different aspects of clinical data

allergies	careplans	claims
claims_transactions	conditions	devices
encounters	medications	observations
imaging_studies	immunizations	organizations
patients	payers	payer_transitions
procedures	supplies	providers

Phase 1

Data Cleaning

The tools and methods used to clean the dataset

Data Cleaning

The cleaning of the data was entirely done using Python libraries, such as pandas and numpy

IDE: Visual Studio Code

Language: Python

Data Cleaning Common Steps

Most columns had a similar issues within them. This is how we dealt with these issues:

- STOP column is often empty so it is filled with the current date instead. (In the used data nulls are replaced with May 1st 2026, the date when the cleaning process originally took place)
- Columns that are duplicated or irrelevant to the analysis were removed.
- Some columns were transformed into a category type for better optimization.
- Replace the null values with proper or stand-in equivalent values.
- Columns that had an incorrect type were corrected

Phase 2 Data Modeling

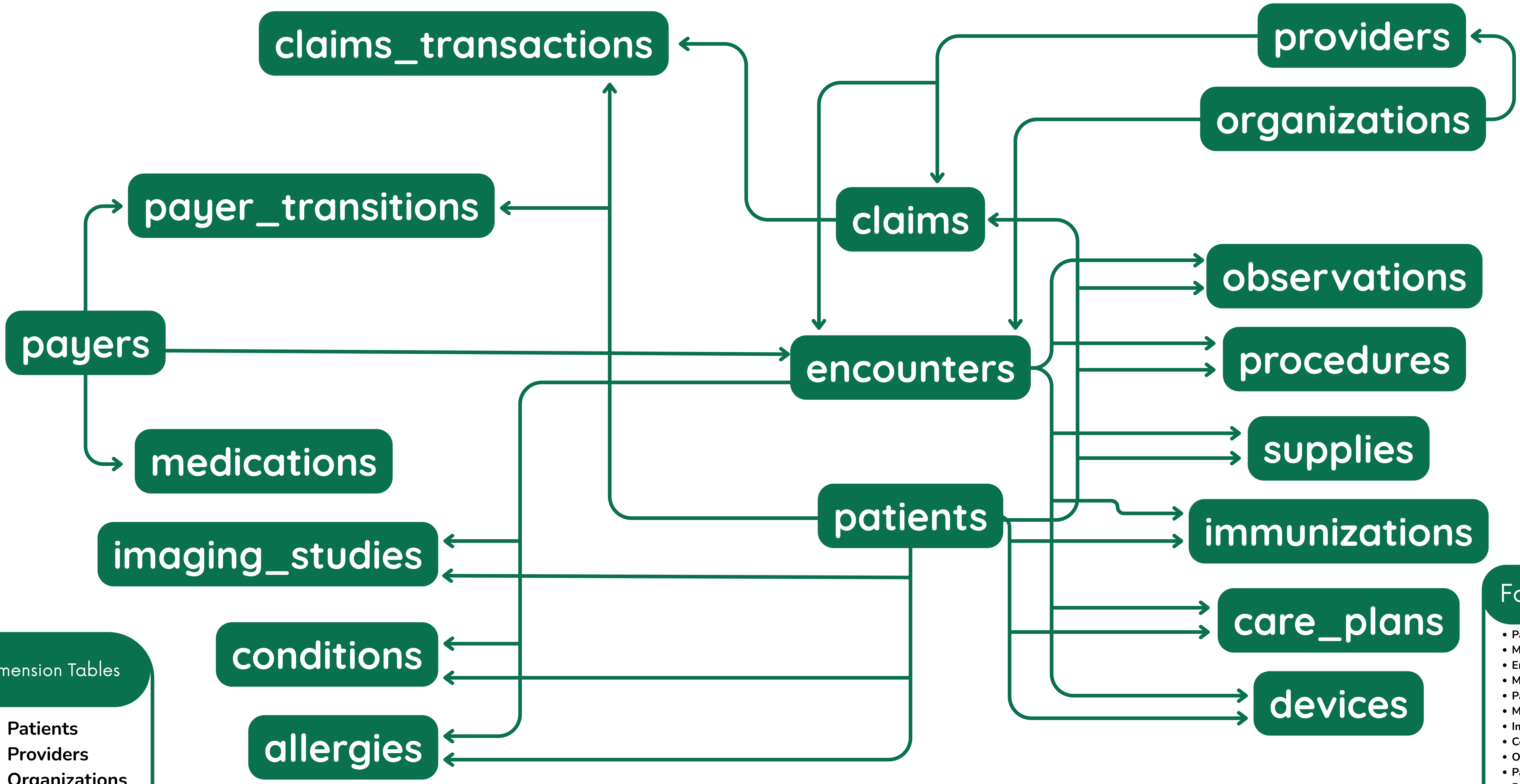
EDA, Relational analysis and measures calculation

Data Modeling

The data model building and measures calculations were made in Microsoft Power BI

Modeling Tool: Microsoft Power BI

Data Modeling ERD



Dimension Tables

- Patients
- Providers
- Organizations
- Payers

Fact Tables

- Patient Allergies
- Medical Claims
- Encounters
- Medications
- Patient Treatment Plans
- Medical Devices
- Imaging Studies
- Conditions
- Observations
- Payer Transitions
- Procedures
- Immunizations
- Financial Transactions

Data Modeling

KPIs and Measures

Patients and Demographics

- Total Patients
- Average Income
- Total Patient Cities
- Unique Patients Visited
- Unique Patients Observed
- Total Healthcare Expenses
- Average Health Coverage
- Unique Patients Imaged
- Unique Patients with Claims
- Unique Patients with Conditions
- Total Patients with Transitions

Clinical & Diagnostics

- Total Encounters
- Avg Cost Per Encounter
- Avg Encounter Duration in Minutes
- Emergency Encounters
- Insurance Coverage Rate
- Total Base Cost
- Total Insurance Covered
- Total Out-of-Pocket Cost
- Total Conditions
- Active Conditions
- Total Unique Diseases
- Avg Conditions Per Encounter
- Avg Conditions Per Patient
- Total Encounters with Conditions
- Total Procedures
- Total Procedure Cost
- Average Cost per Procedure
- Avg Procedures per Patient
- Total Unique Patients
- Procedures Cost LY
- Total Observations Count
- Vital Signs Observations
- Laboratory Observations
- Laboratory Observations Percentage
- Avg Systolic New
- Avg Diastolic New
- Average Body Weight
- Total Allergies
- Unique Patients with Allergies
- Allergy Density
- Average BMI New
- Total Imaging Studies
- Avg Studies Per Patient
- CT Scans Count
- CT Scan Rate
- Thoracic Imaging Count
- Modality Contribution
- Severe Allergies Count
- Severe Allergies Rate
- Moderate Allergies Count
- Current Month Allergies
- Allergies Previous Month

Data Modeling

KPIs and Measures

Financial & Costs

- Number of Transactions
- Transactions Sum
- Transfers Sum
- Total Immunization Base Cost
- Average Immunization Base Cost
- Total Dispenses
- Total Medications Cost
- Average Cost Per Medication
- Total Payer Coverage
- Medication Insurance Coverage Rate
- Out-of-Pocket Rate
- Total Claims Count
- Insured Claims Count
- Insurance Coverage Claims Rate
- Claims Distribution
- Total Supplies
- Total Supply Records

Providers & Organizations

- Total Providers
- Total Provider Encounters
- Total Cities Covered
- Male Encounters
- Female Encounters
- Average Provider Encounters
- Total Organizations
- Total Cities
- Total Utilization
- Avg Utilization
- Max Utilization
- Unique Providers

Payers & Coverage

- Total Revenue
- Total Payers Count
- Total Amount Covered
- Total Amount Uncovered
- Total Uncovered Expenses
- Total Covered Encounters
- Total Covered Medications
- Total Unique Customers
- Total Transitions
- Active Transitions
- Self Paid Transitions
- Transitions This Year
- Patients with Multiple Transitions
- Avg Coverage Duration (Days)
- Uninsured Rate

Phase 3 Visualization

Dashboard design and Finalization

Visualization

The Visualization process is also done using Microsoft Power BI

Visualization Tool : Microsoft Power BI

SYNTHEA Data Analysis Dashboard

Overview | Clinical Analysis | Financial Analysis

116
Total Patients

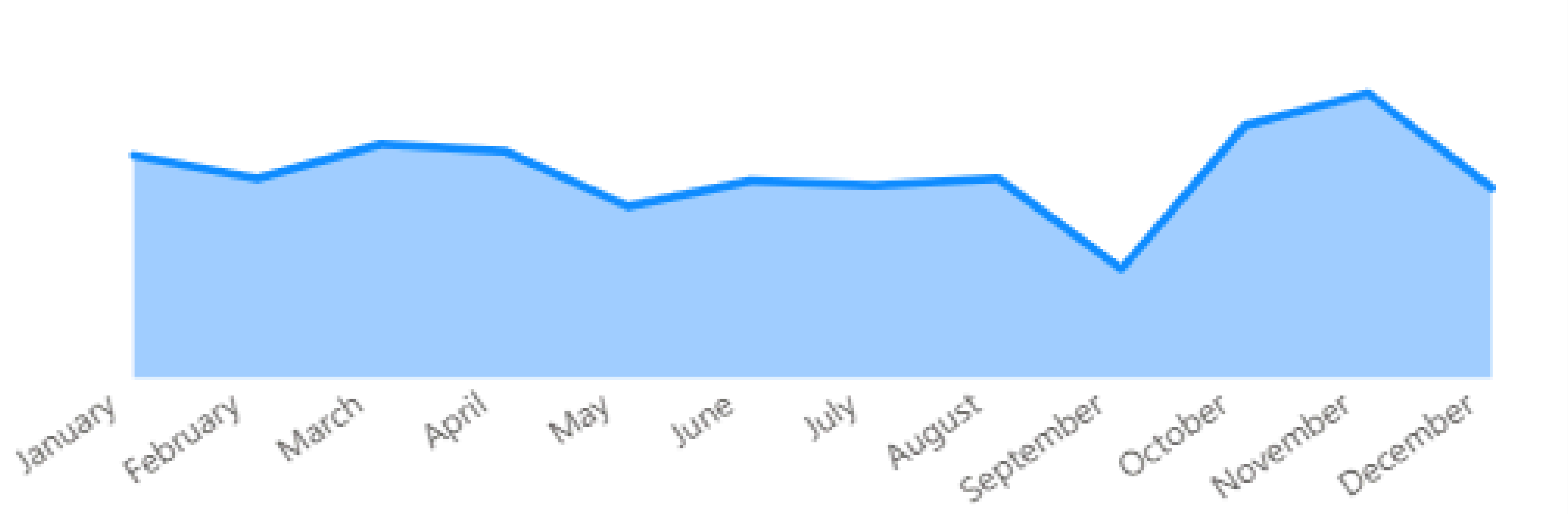
6K
Total Encounters

$31.74M
Transactions Sum

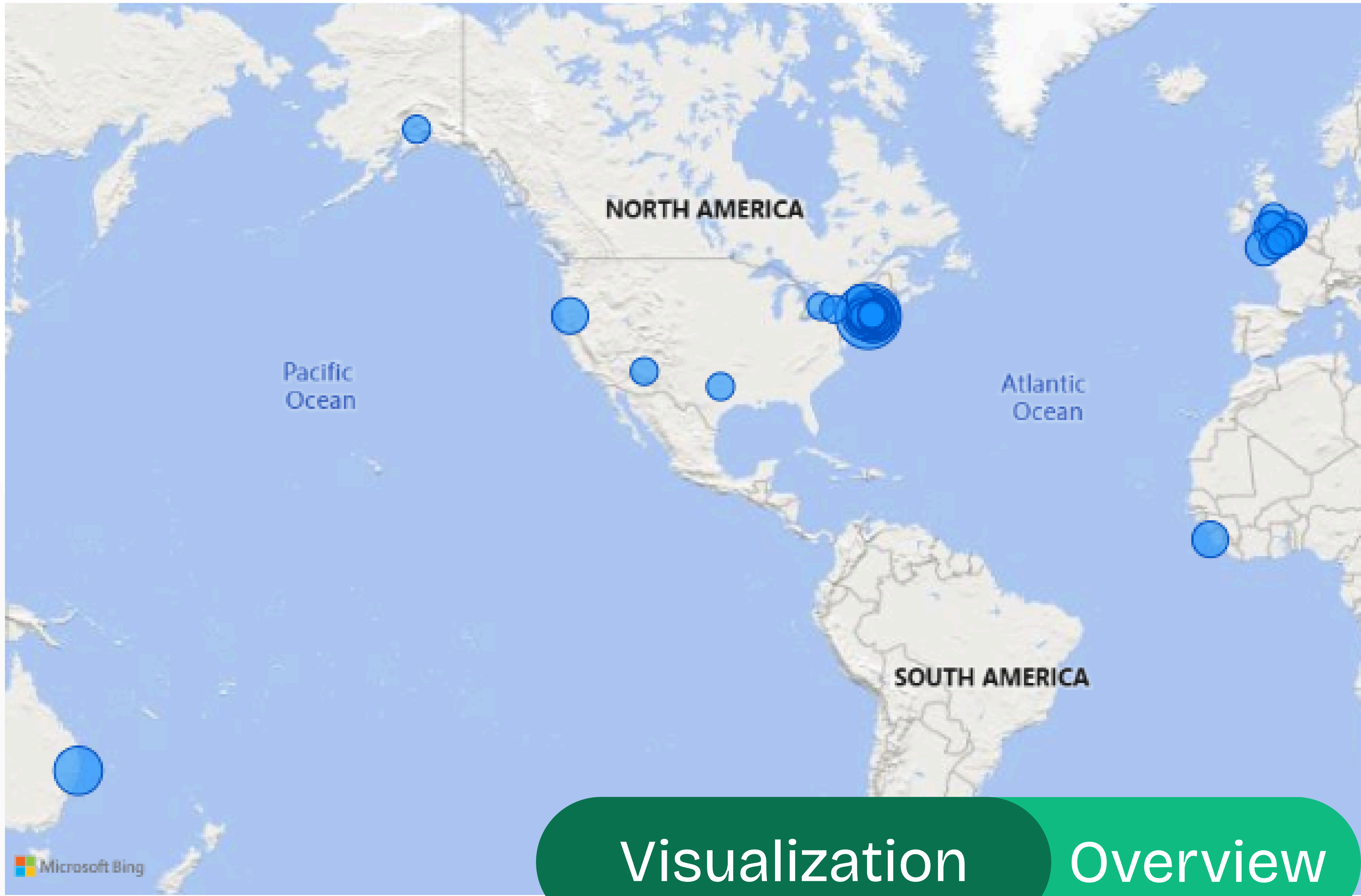
$4.70M
Transfers Sum

Encounter Date
1/15/1944 3/5/2026

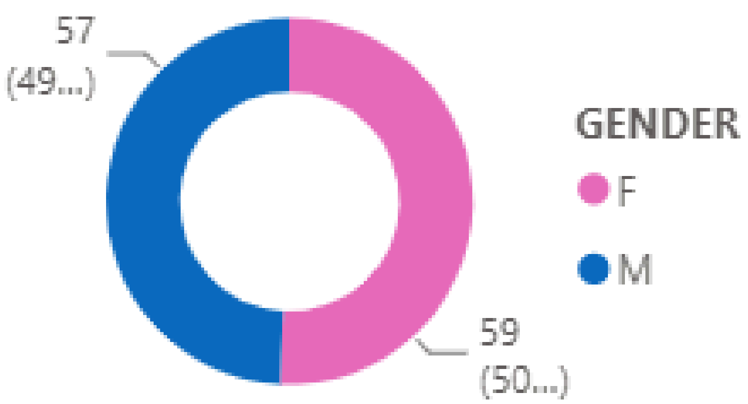
Monthly Activity



Patient Locations



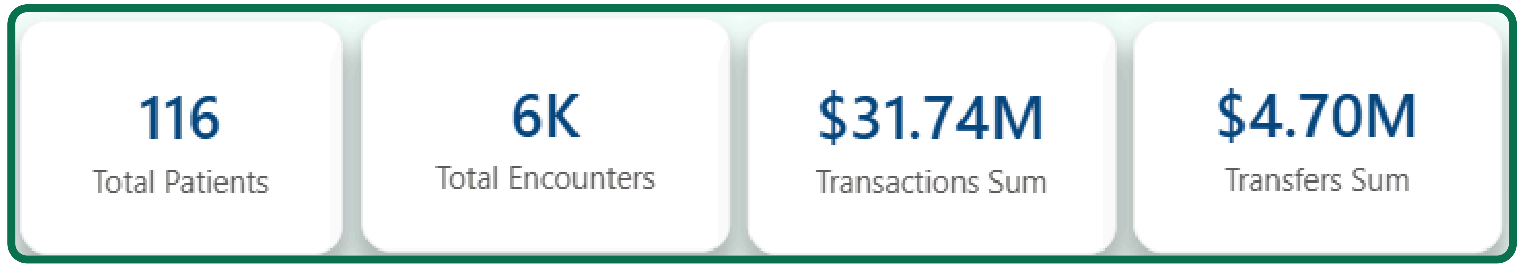
Patient Gender Distribution



Allergy Density Across the Years

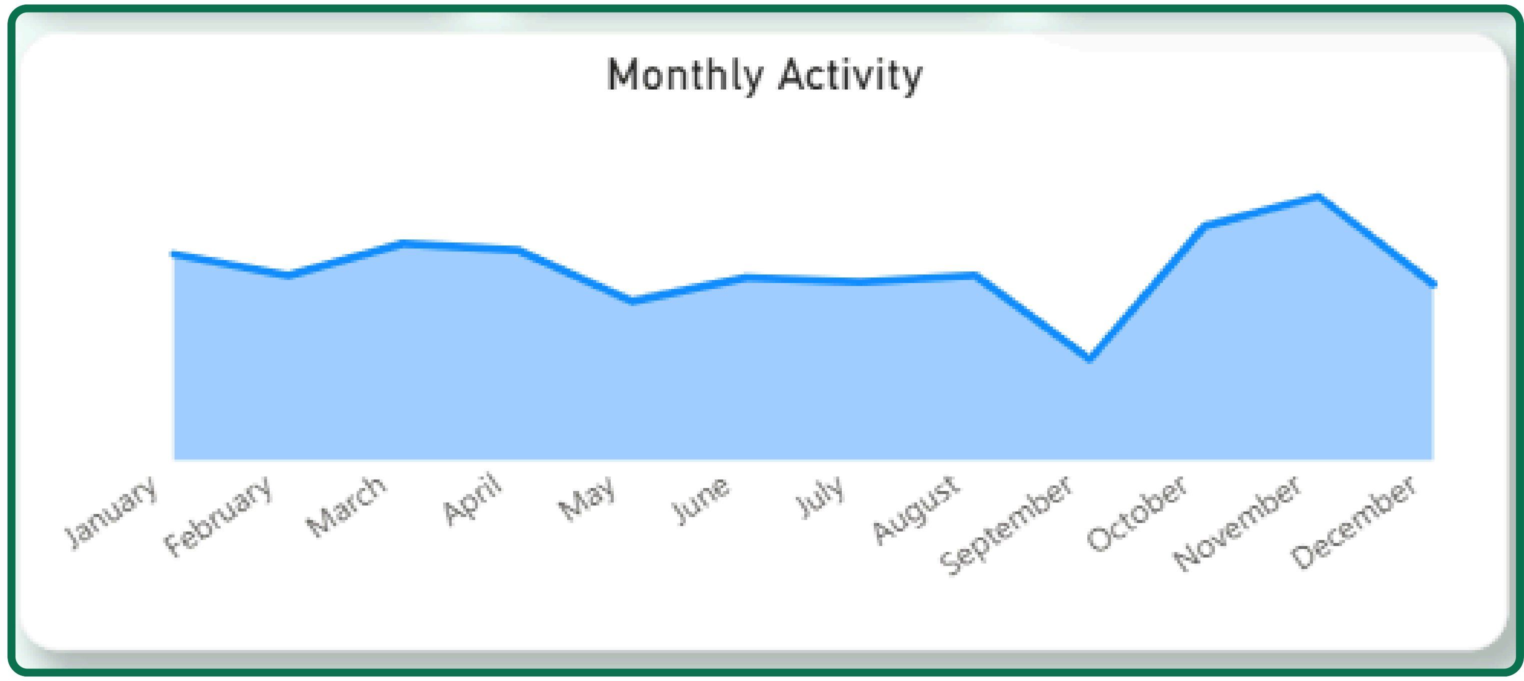
Visualization Overview

Visualization Overview

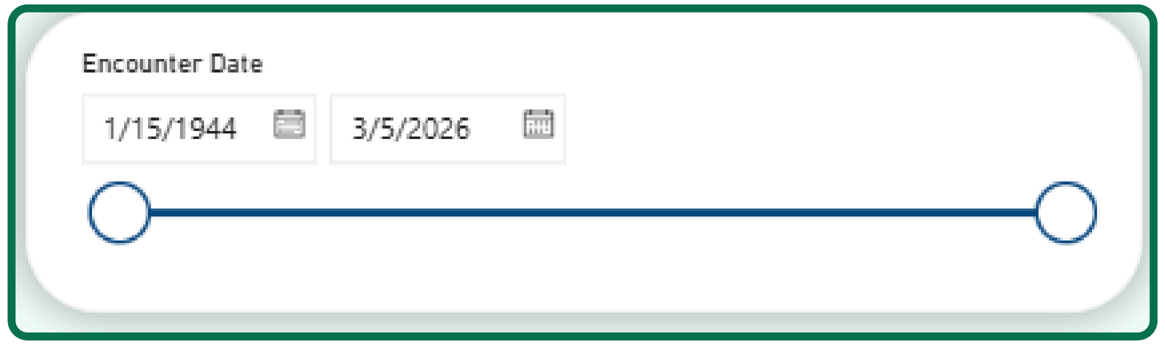


Basic measures

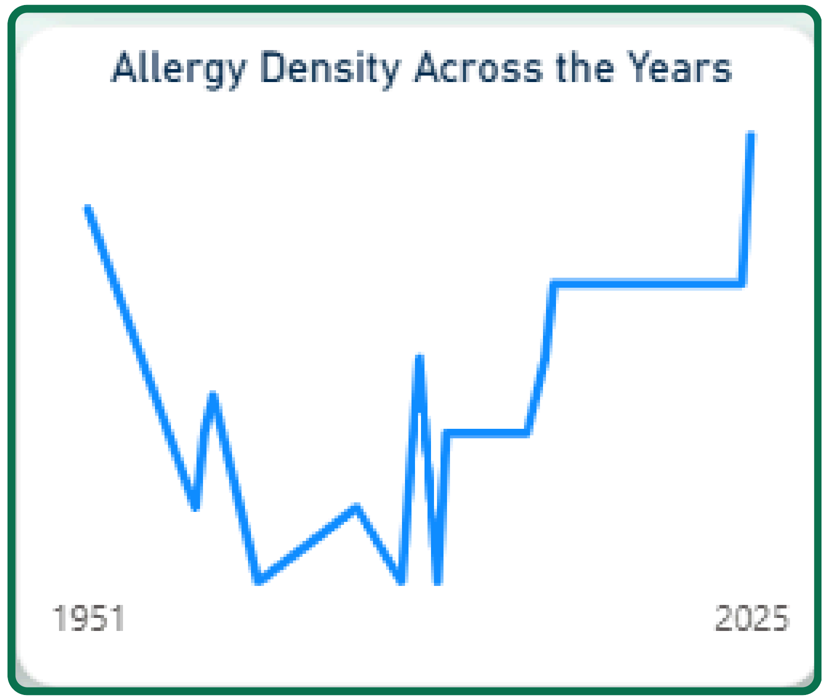
Monthly activity based on encounters



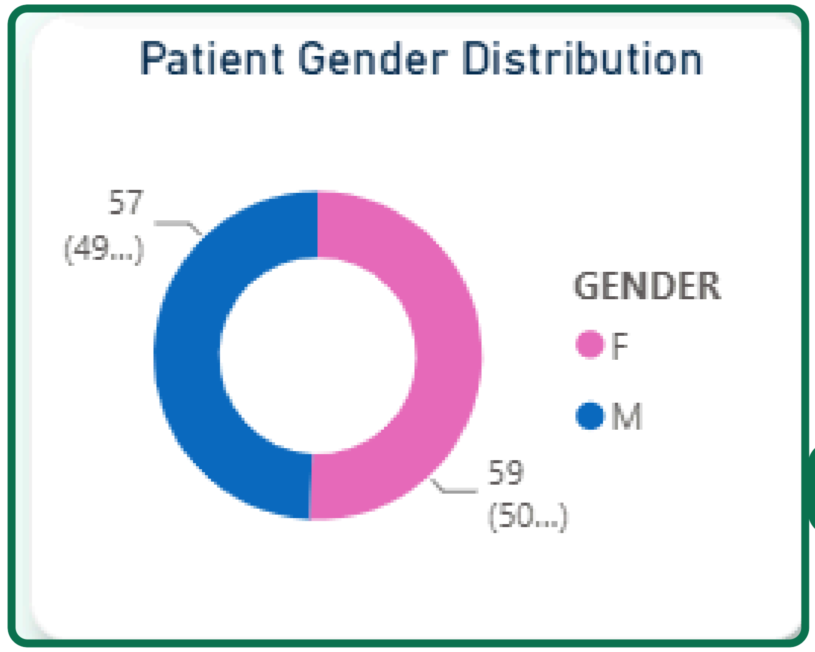
Navigation Bar



A slicer that changes time based on encounter start date

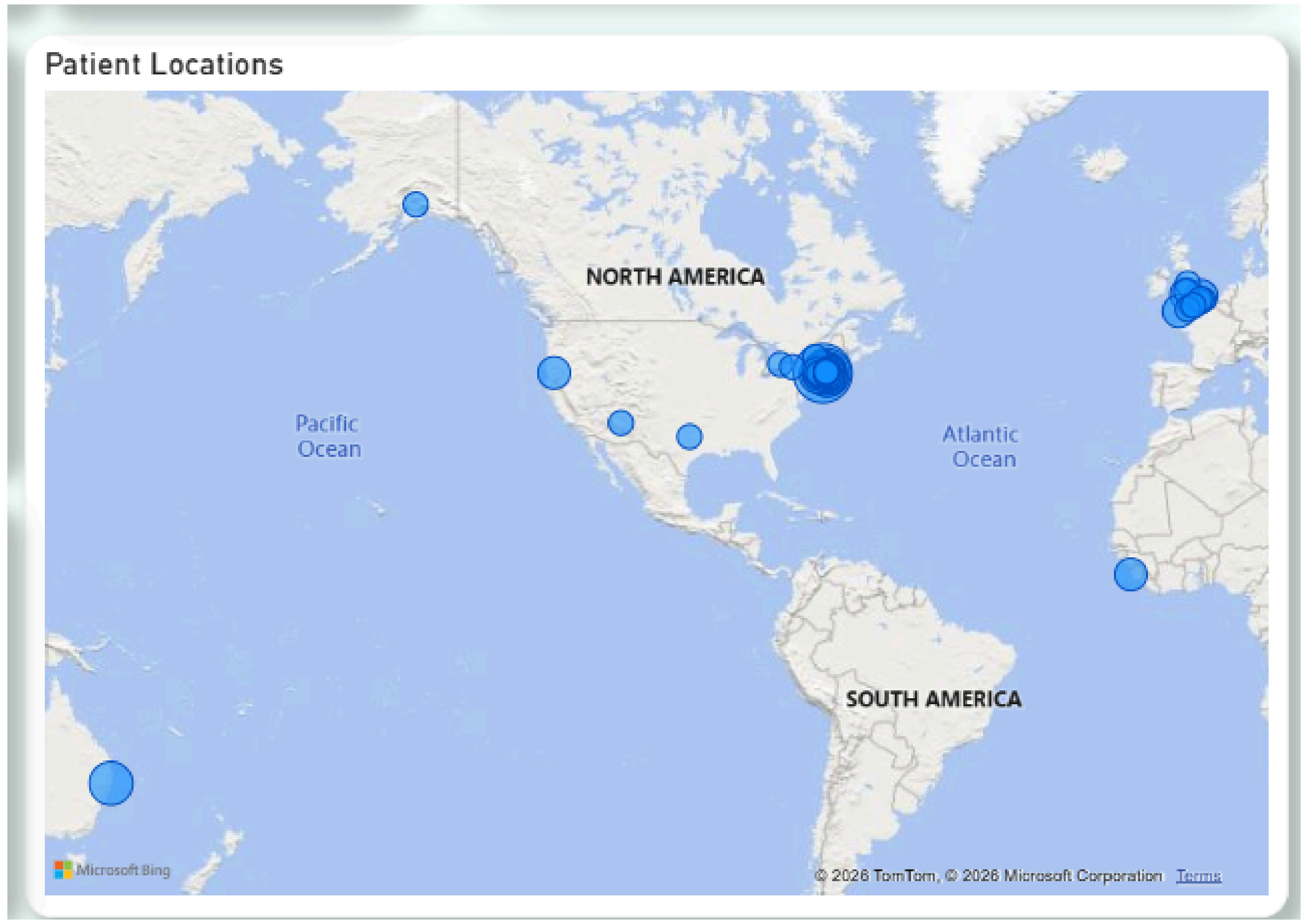


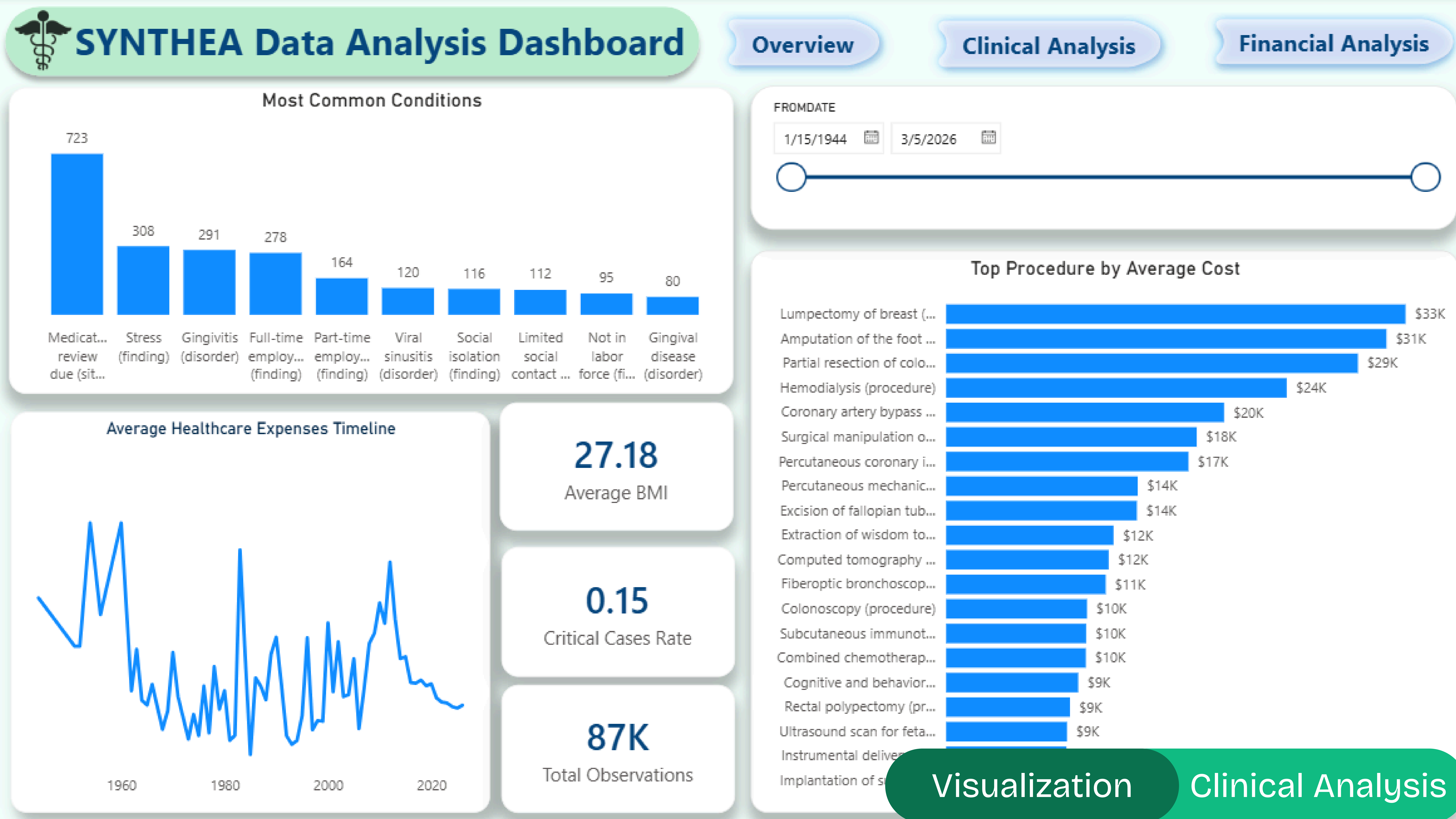
A line chart showing the change in allergies per patient over the years



A donut chart showing the distribution of patients based on gender

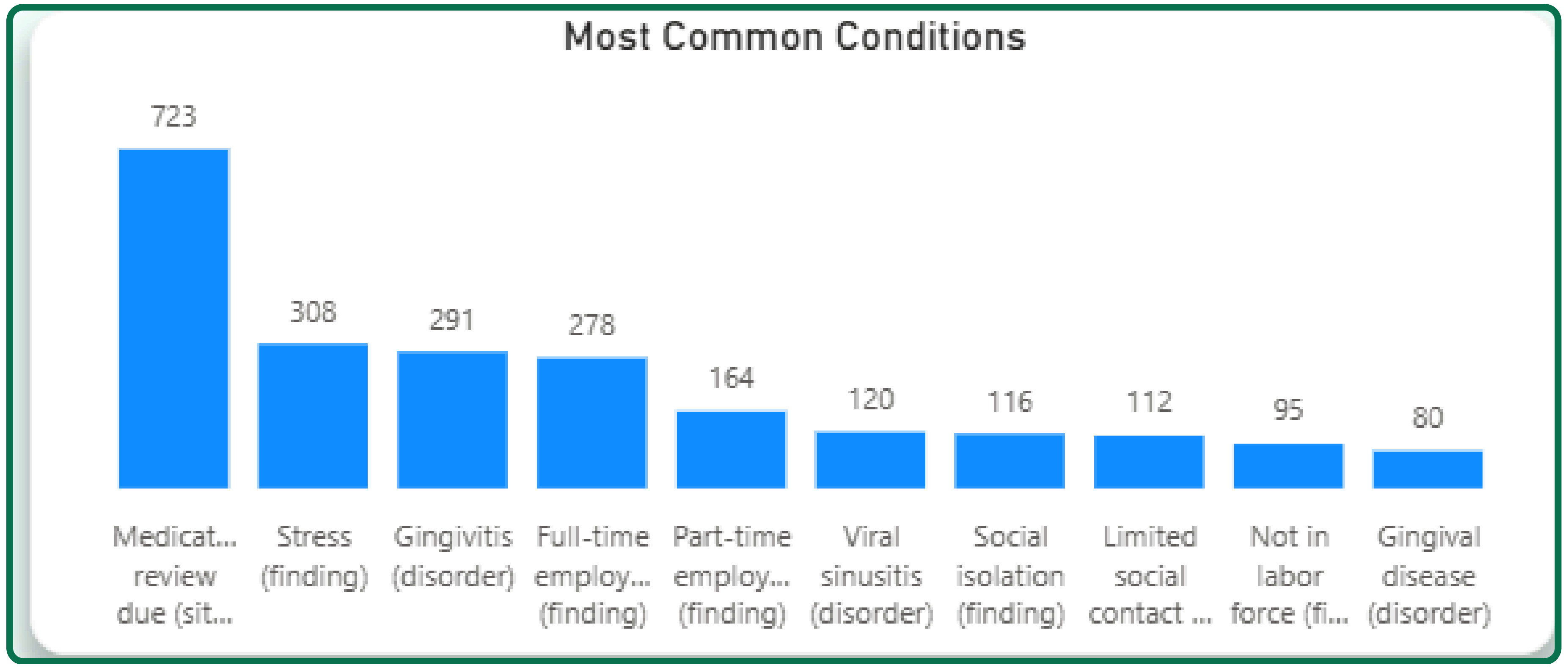
Patient distribution across the world



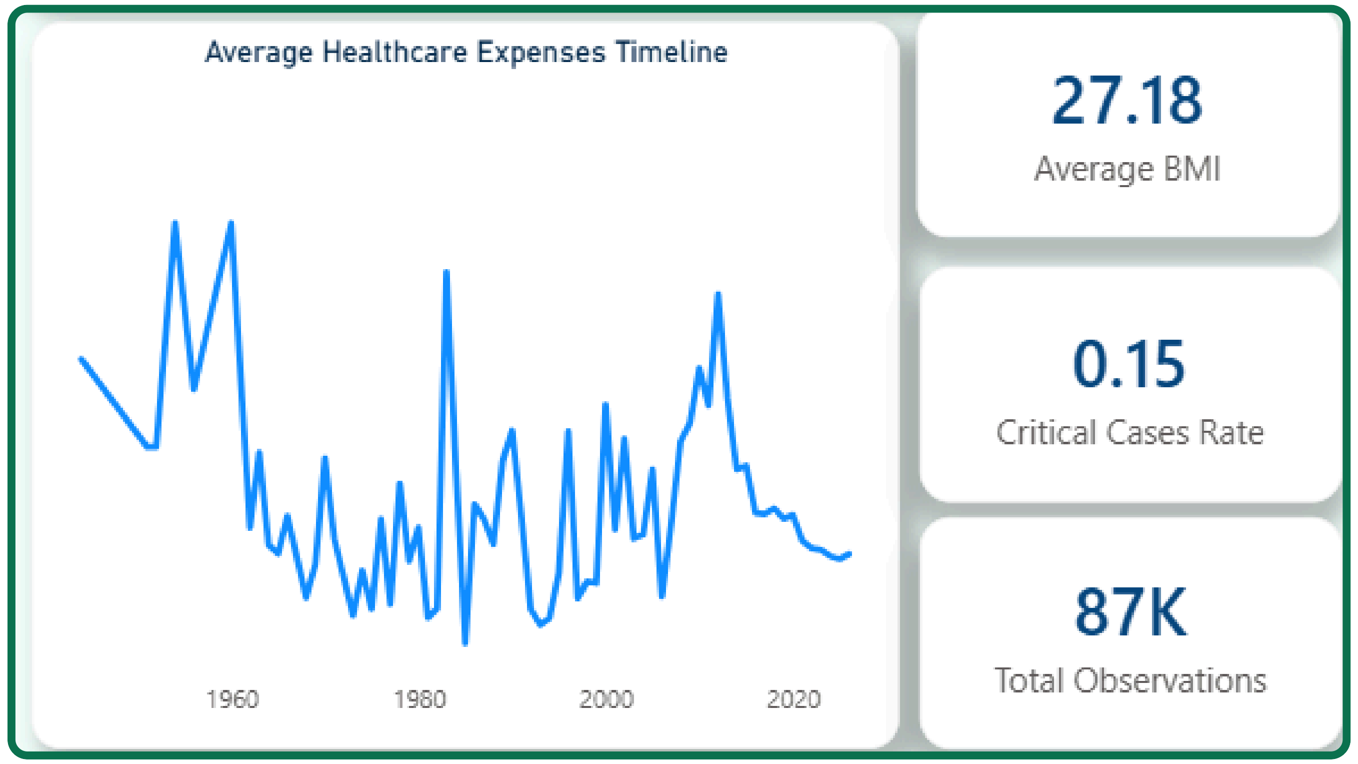


SYNTHEA Data Analysis Dashboard
Overview
Clinical Analysis
Financial Analysis
Most Common Conditions
723
308
291
278
164
120
116
112
95
80
Medicat... review due (sit...
Stress (finding)
Gingivitis (disorder)
Full-time employ... (finding)
Part-time employ... (finding)
Viral sinusitis (disorder)
Social isolation (finding)
Limited social contact ...
Not in labor force (fi...
Gingival disease (disorder)
FROMDATE
1/15/1944
3/5/2026
Top Procedure by Average Cost
Lumpectomy of breast (... $33K
Amputation of the foot ... $31K
Partial resection of colo... $29K
Hemodialysis (procedure) $24K
Coronary artery bypass ... $20K
Surgical manipulation o... $18K
Percutaneous coronary i... $17K
Percutaneous mechanic... $14K
Excision of fallopian tub... $14K
Extraction of wisdom to... $12K
Computed tomography ... $12K
Fiberoptic bronchoscop... $11K
Colonoscopy (procedure) $10K
Subcutaneous immunot... $10K
Combined chemotherap... $10K
Cognitive and behavior... $9K
Rectal polypectomy (pr... $9K
Ultrasound scan for feta... $9K
Average Healthcare Expenses Timeline
1960
1980
2000
2020
27.18
Average BMI
0.15
Critical Cases Rate
87K
Total Observations
Visualization
Clinical Analysis

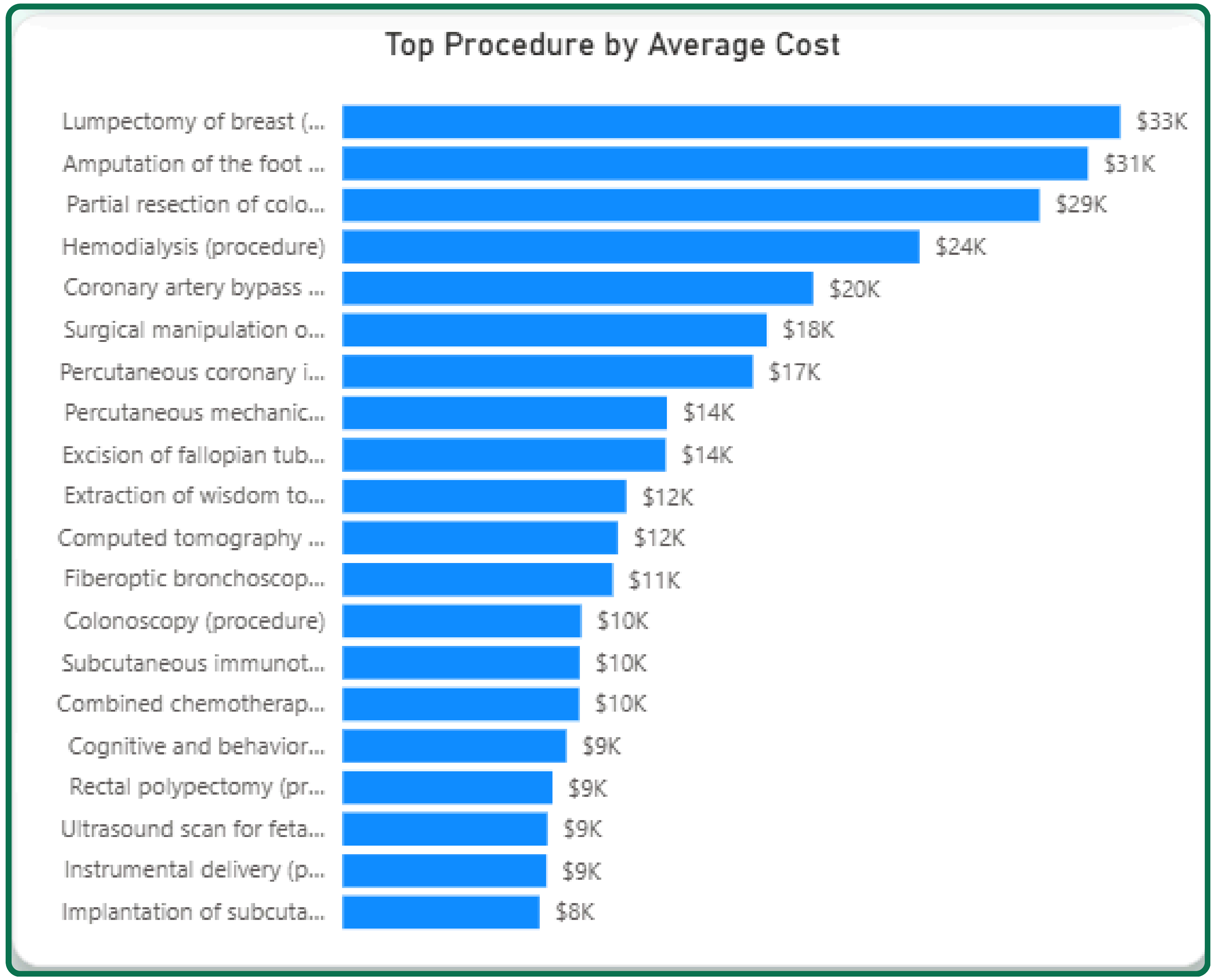
Visualization Clinical Analysis



Clustered Column Chart of the Top 10 common conditions among patients



A Line Chart showcasing The evolution of average healthcare expenses across the years alongside some extra major measures



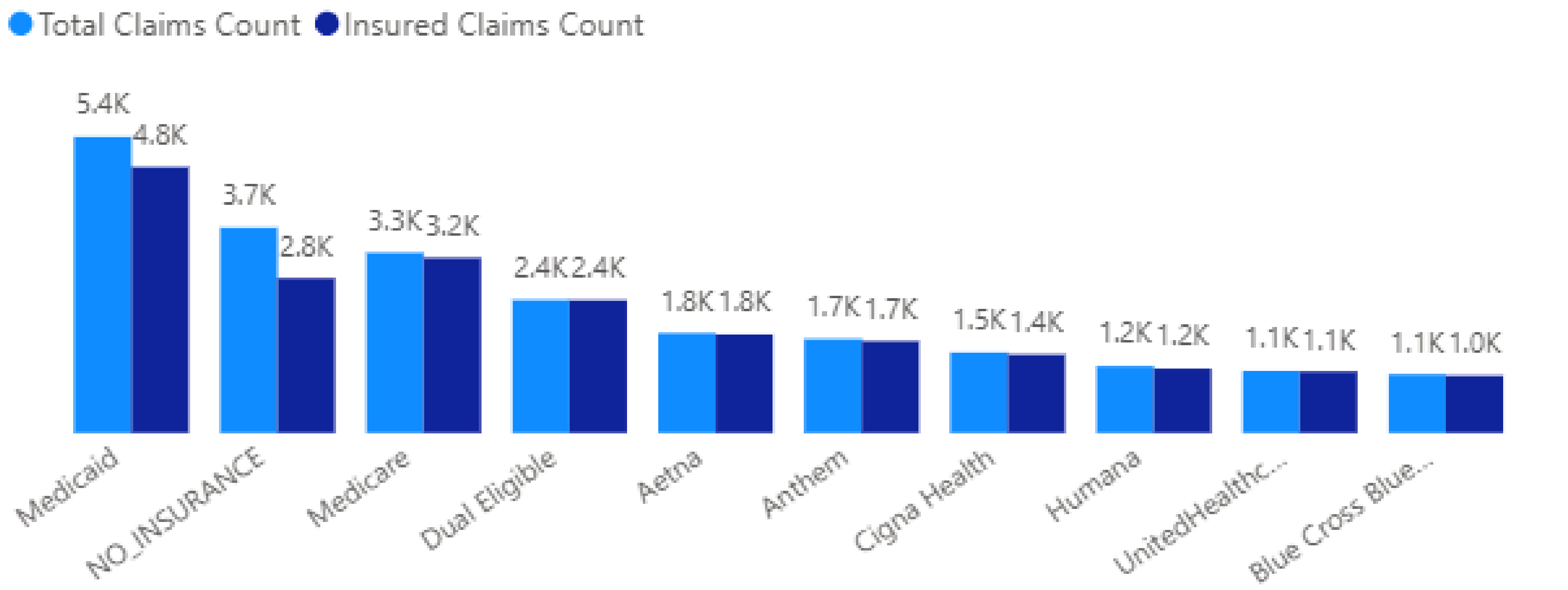
Clustered Bar Chart of the Top 20 Procedures by their average cost

Slicer changing time based on procedure start date

SYNTHEA Data Analysis Dashboard

Overview | Clinical Analysis | Financial Analysis

Distribution of Payers by Insured vs Uninsured Claims



Payer Coverage Information

NAME	Total Transitions	Uninsured Rate	Avg Coverage Duration (Days)
Medicare	1904	0.23	365.02
NO_INSURANCE	1694	0.56	365.01
Medicaid	1418	0.15	364.43
Cigna Health	1209	0.22	364.01
Aetna	1088	0.17	364.20
Humana	1057	0.25	365.21
Anthem	768	0.29	363.80
Blue Cross Blue Shield	630	0.25	365.21
UnitedHealthcare	359	0.33	365.23
[illegible]	[illegible]	[illegible]	[illegible]
Total	**4810**	**0.25**	**364.84**

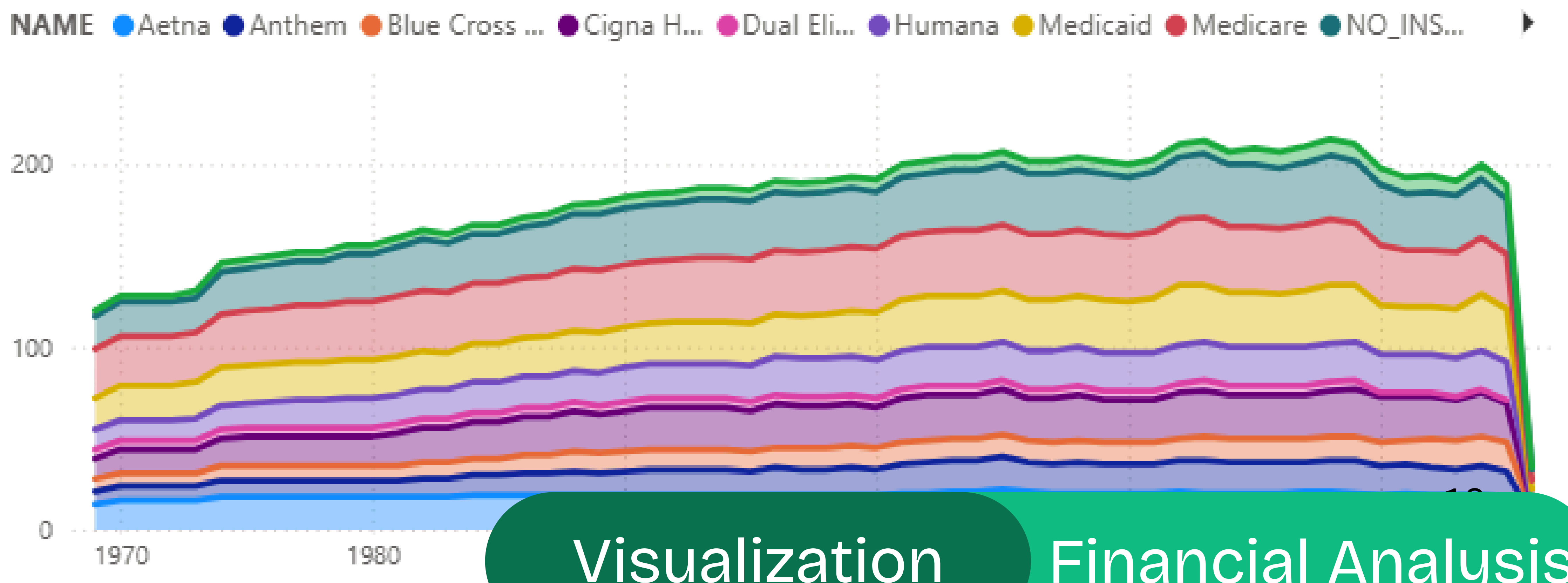
5K
Total Transitions

0.25
Uninsured Rate

$1.34K
Average Procedure Cost

$27.19M
Total Procedure Cost

Trends by Payer

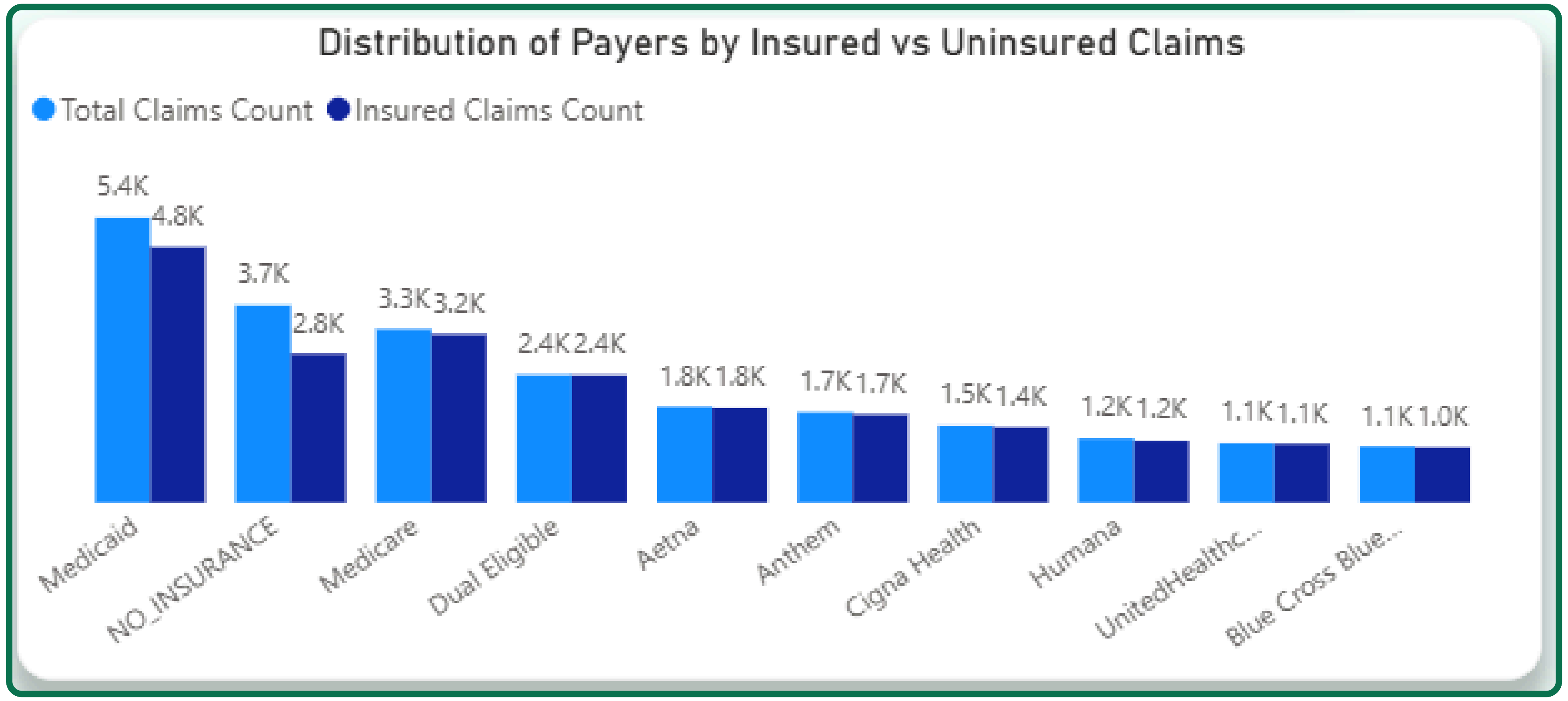


Visualization

Financial Analysis

Visualization

Financial Analysis



Clustered column chart showcasing the difference between insured and uninsured claims for every payer

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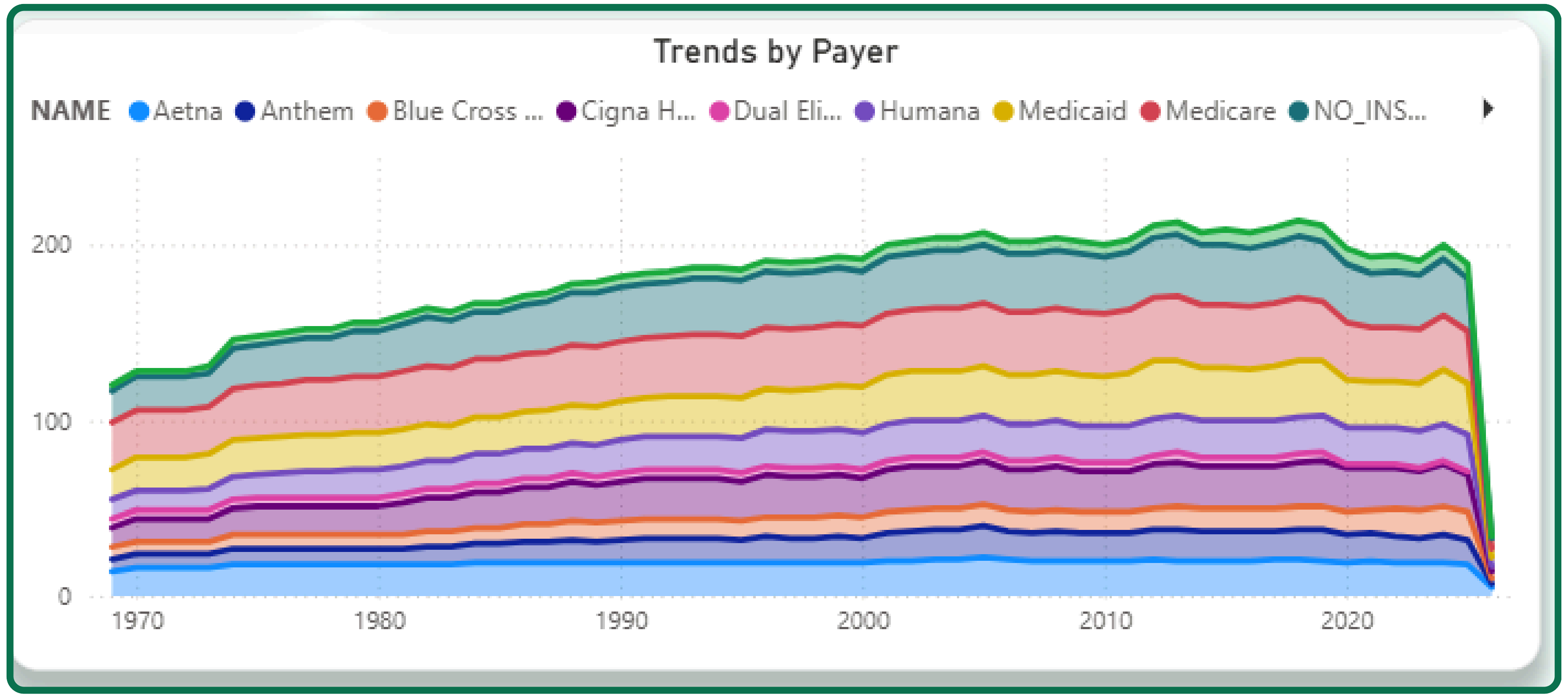
A table showing every payer's total transitions, uninsured rate and average coverage duration



Stacked area chart showing number of transitions across different payers

5K Total Transitions	**0.25** Uninsured Rate
$1.34K Average Procedure Cost	**$27.19M** Total Procedure Cost

Financial measures

Insights

- **Lupectomy of breast and foot amputations are the most common procedures.**
- **Average healthcare expenses keep going up and down, but have decreased over all.**
- **Most patients come from the United States and England.**
- **Activity is at its peak during November and at its lowest during September, which is interesting considering that the difference between them is not that large.**

Insights

- Allergy density saw a huge drop from 6.00 to 2.00 between 1951 and 1963, followed by a period of constant spiking and dropping until reaching a period of stability between 2003 and 2024 followed by an all-time high density of 7.00 during 2025
- Female patients are generally more common than male patients (52.21% against 47.79%)
- UnitedHealthCare has the highest average coverage duration followed by Blue Cross Blue Shield and Humana
- Medicare has the highest amount of transitions

That is all!

Thank you for your attention.

See you again soon!

Best Regards,

The Outliers Team

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Data Curation: Mohammad Abdulhakim
Data Modeling: Hoda Elkady
Visualization: Abdulrahman Hisham